The Big Bang as Energy-Triggered Condensation of an Infinite Two-Spinor Complexified 4D Vacuum into a $3+1$D Spacetime Bubble: A Cosmological Toy Model

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Abstract

Working within a two-spinor Complexified 4D Vacuum ($\mathbb{C}^4 = \mathbb{C}^2_L \oplus \mathbb{C}^2_R$) [5, 6], we formulate a self-contained cosmological toy model in which the **Big Bang is not a $3+1$D geometric point singularity of infinite density, but a localized, energy-triggered phase transition (condensate bubble nucleation) inside an infinite, subcritical, scale-free $\mathbb{C}^4$ vacuum**. In its ambient pre-Bang state, the infinite $\mathbb{C}^4$ bulk is subcritical ($g_\Sigma < 1 \implies m^2_{\text{eff},0} > 0$), possessing neither a $3+1$D metric ($\langle \Sigma_{\alpha\beta} \rangle = 0$) nor a causal arrow of time ($\langle H_{\alpha\bar{\beta}} \rangle = 0$), and residing entirely in the time-symmetric Bargmann–Hall–Wightman Extended Tube ($W^+ = W^-$) [2] with zero bare cosmological constant ($\rho_{\text{vac}}(\text{empty } \mathbb{C}^4) = 0$). However, $\mathbb{C}^4$ supports propagating holomorphic spinor ripples $\Psi_\alpha(Z)$. When a large-scale constructive interference (focal caustic) of holomorphic ripples concentrates invariant energy density $\mathcal{E}_{\text{loc}}(Z)$ above the critical nucleation threshold $\mathcal{E}_{\text{crit}} = (m^2_{\text{eff},0}/\alpha_{\text{E}})^2$, the effective quadratic curvature flips negative, nucleating an expanding $3+1$D condensate bubble [3, 4]. As the bubble expands and cools, it steps through a three-stage symmetry-breaking thermal history: (1) **Birth of the Arrow of Time** ($\langle H_{\alpha\bar{\beta}} \rangle$ selecting signature $(+,+,-,-)$ and partitioning $\mathbb{PT}^+$ from $\mathbb{PT}^-$ [5]); (2) **Planck/Inflationary Slice Condensation & Reheating into the Radiation Era** ($\langle \Sigma_{\alpha\beta} \rangle = \rho_{\text{grav}}(t) e^{i\chi} I_{\alpha\beta}$ [6], whose latent heat populates the newly formed $3+1$D light-cone surface $r_{\text{Bloch}} = 1$ with massless chiral ripples obeying $w = P/\varepsilon = \frac{1}{3}\langle r^2_{\text{Bloch}} \rangle = 1/3$); and (3) **Electroweak Higgs Condensation into the Matter Era** ($T_{\text{EW}} \approx 160$ GeV [7], tethering counter-propagating radiation ripples into massive Dirac standing waves $E^2 = (mc^2)^2 + (pc)^2$ [8, 9] inside the Bloch ball $r_{\text{Bloch}} = v/c < 1$, driving $w(T) \to 0$). We show how this framework naturally addresses the initial singularity [10], the origin of the arrow of time, the CMB horizon problem (via pre-light-cone holomorphic equilibration in the $\mathbb{C}^4$ Extended Tube), and the cosmological constant problem (via the expanding bubble's surface-area × zero-point energy density $\rho_\Lambda = \frac{A_{\text{surface}}}{V_{\text{bulk}}} \frac{\frac{1}{2}\hbar H_0}{\ell^2_{\text{Pl}}} \sim (\hbar H_0 M_{\text{Pl}})^2$). Finally, we contrast the model with standard Twistor Theory and Penrose's Objective Reduction [31, 33], and derive three falsifiable observational signatures for CMB low-ℓ multipoles, primordial gravitational wave parity, and weakening dynamical dark energy ($w_0 > -1, w_a < 0$) [26, 27].

Keywords: Big Bang Cosmology, Complexified Minkowski Space, Twistor Theory, Vacuum Phase Transitions, Bubble Nucleation, Arrow of Time, Horizon Problem, Induced Gravity, Dirac Standing Waves, Dynamical Dark Energy

1 Introduction & Cosmological Motivation

Standard Friedmann–Lemaître–Robertson–Walker (FLRW) cosmology, combined with General Relativity (GR) and the Standard Model of particle physics, successfully describes the thermal evolution of the universe from a fraction of a second after the Big Bang to the present epoch [11]. Yet when extrapolated back to $t \to 0$, the standard $3+1$D paradigm confronts four foundational puzzles:

1. **The Initial $t = 0$ Singularity Problem:** By the Hawking–Penrose singularity theorems [12], evolving a classical $3 + 1$D Lorentzian metric $g_{\mu\nu}$ backward in time under standard energy conditions forces the scale factor $a(t) \to 0$ and the Riemann curvature invariants to diverge ($R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \to \infty$), signaling the breakdown of $3 + 1$D spacetime geometry itself.

2. **The Origin of the Arrow of Time & Low-Entropy Initial State:** The microscopic laws of QED, QCD, and GR are time-reversal symmetric, yet the universe began with an extraordinarily low gravitational entropy (vanishing Weyl curvature, $C_{\mu\nu\rho\sigma} \to 0$ [10]) and a globally directed thermodynamic and causal arrow of time.

3. **The Horizon Problem:** Regions of the Cosmic Microwave Background (CMB) separated by more than $\sim 2^\circ$ on the sky were spacelike-separated (outside each other's $3+1$D past light cones) at the time of recombination, yet their temperatures and phase fluctuations are equilibrated to $\Delta T/T \sim 10^{-5}$ [13]. While scalar-field inflation [14, 15] addresses this by postulating an ad-hoc inflaton potential, the microscopic origin of the inflaton and its initial pre-inflationary homogeneity remain open questions.

4. **The Cosmological Constant Catastrophe:** If the $3 + 1$D vacuum is treated as a uniform, infinite ground state with Planck-scale zero-point energy density $\rho_{\text{vac}} \sim M_{\text{Pl}}^4$, the resulting cosmological constant exceeds the observed late-time dark energy density $\rho_\Lambda^{\text{obs}} \approx (2.3 \times 10^{-3}\ \text{eV})^4$ by ~ 120 orders of magnitude [16, 4].

Core Thesis of This Paper

In a two-spinor **Complexified 4D Vacuum ($\mathbb{C}^4 = \mathbb{C}^2_L \oplus \mathbb{C}^2_R$)** [5, 6], $3+1$D relativistic kinematics, the Kaluza–Klein $U(1)$ electromagnetic fiber [17, 18, 19], the chiral Dirac equation [8, 9], and outside-the-light-cone quantum correlations [20, 21] emerge as low-energy projections of spinor pairing. Crucially, the empty $\mathbb{C}^4$ vacuum need not be unstable on its own: in the subcritical Nambu–Jona-Lasinio (NJL) [22] regime, empty $\mathbb{C}^4$ remains uncondensed ($\langle\rho\rangle_{\text{empty } \mathbb{C}^4} = 0$), while localized concentrations of holomorphic 4D ripple energy ($\mathcal{E}_{\text{loc}} \geq \mathcal{E}_{\text{crit}}$) nucleate $3 + 1$D condensate droplets.

In this paper, we formulate that mechanism at the cosmological scale: **we model the Big Bang as the nucleation of an expanding $3+1$D condensate bubble inside an infinite, subcritical $\mathbb{C}^4$ vacuum triggered by a massive constructive-interference concentration (focal caustic) of holomorphic 4D ripples.**

2 The Pre-Bang State: The Infinite Subcritical $\mathbb{C}^4$ Vacuum

2.1 Ambient $SL(4,\mathbb{C})$ Symmetry and Absence of a $3+1$D Metric

Consider an infinite, 4-dimensional complex manifold $\mathbb{C}^4 = \mathbb{C}^2_L \oplus \mathbb{C}^2_R$ parameterized by holomorphic twistor-spinor coordinates [5, 6, 23]:

$$Z^\alpha = (\omega^A,\, \pi_{A'}) \in \mathbb{C}^4 \qquad (\alpha = 0,1,2,3; \quad A = 0,1; \quad A' = 0',1'). \tag{1}$$

Prior to spontaneous symmetry breaking, the ambient symmetry group of $\mathbb{C}^4$ is the complex special linear group $SL(4,\mathbb{C})$, and the microscopic dynamics is governed purely by massless twistor-spinor fields $\Psi_\alpha(Z) = (\xi_A(Z), \eta^{A'}(Z))$ with four-spinor contact interactions [22]:

$$\mathcal{L}_{\text{micro}}(\Psi) = i\hbar c\, \bar\Psi_{\bar\alpha}\mathcal{D}^{\bar\alpha\beta}\Psi_\beta + \frac{G_H}{2}\left(\bar\Psi_{\bar\alpha}\Psi_\beta\right)\left(\bar\Psi_{\bar\beta}\Psi_\alpha\right) + \frac{G_\Sigma}{4}\left|\Psi_{[\alpha}\tilde\Psi_{\beta]}\right|^2 + G_{\text{EW}}\left(\xi^A\eta_A\right)\left(\bar\xi_{A'}\bar\eta^{A'}\right). \tag{2}$$

Introducing the collective sesquilinear slice-order parameter $H_{\alpha\bar\beta} \sim \langle\Psi_\alpha\bar\Psi_{\bar\beta}\rangle$ and the antisymmetric bi-twistor condensate $\Sigma_{\alpha\beta} = \rho\, e^{i\chi} I_{\alpha\beta} \sim \langle\Psi_{[\alpha}\tilde\Psi_{\beta]}\rangle$ via Hubbard–Stratonovich auxiliary fields and integrating out 1-loop spinor fluctuations up to the cutoff Λ_{UV} yields the empty-vacuum quadratic stiffness [22, 24]:

$$m^2_{\text{eff},0} \equiv \frac{1}{G_\Sigma} - \frac{N_f g_{\text{Y}}^2 \Lambda^2_{\text{UV}}}{8\pi^2} = \frac{1-g_\Sigma}{G_\Sigma}, \qquad g_\Sigma \equiv \frac{N_f g_{\text{Y}}^2 \Lambda^2_{\text{UV}}}{8\pi^2}\, G_\Sigma. \tag{3}$$

Our foundational cosmological postulate is that the infinite ambient $\mathbb{C}^4$ vacuum lies in the **subcritical regime**:

$$g_\Sigma < 1 \quad\Longrightarrow\quad m^2_{\text{eff},0} > 0 \quad\Longrightarrow\quad \langle H_{\alpha\bar\beta}\rangle_{\text{bulk}} = 0, \quad \langle\Sigma_{\alpha\beta}\rangle_{\text{bulk}} = 0. \tag{4}$$

This subcritical pre-Bang vacuum exhibits three remarkable physical properties:

1. **Primordial Spinor Vacuum Ontology (Gravity Does Not Pre-Exist the Condensation):** In the ambient $\mathbb{C}^4$ bulk, the *only* fundamental pre-existing quantum field is the primordial spinor field $\Psi_\alpha(Z)$. The auxiliary field $\Sigma_{\alpha\beta}(Z) \equiv \bar\Psi_{[\alpha}(Z)\Psi_{\beta]}(Z)$ is merely a Hubbard–Stratonovich collective shorthand for spinor pairing (analogous to Cooper pairs in BCS superconductivity). Because $\langle\Sigma_{\alpha\beta}\rangle_{\text{bulk}} = 0$, the gravitational field and the $3+1$D metric $ds^2_{\text{4D}} \propto |\langle\Sigma_{\alpha\beta}\rangle Z_1^\alpha Z_2^\beta|^{-2}$ do not exist prior to condensation; empty $\mathbb{C}^4$ has no rulers, no clocks, and no preferred real $(3,1)$ slice. Gravity is born at the condensation event itself as the collective elastic Goldstone/hydrodynamic mode of the phase-locked spinors.

2. **Time-Symmetric Extended Tube (No Arrow of Time):** Because $\langle H_{\alpha\bar\beta}\rangle_{\text{bulk}} = 0$, the positive-frequency forward tube $\mathbb{PT}^+$ and negative-frequency backward tube $\mathbb{PT}^-$ are not yet separated. The entire ambient $\mathbb{C}^4$ bulk exists in the time-symmetric Bargmann–Hall–Wightman **Extended Tube** [2], where forward and backward correlators are identical $(W^+(Z_1,Z_2) = W^-(Z_1,Z_2))$.

3. **Zero Bare Cosmological Constant in the Bulk:** Because the order parameters vanish identically in the uncondensed $\mathbb{C}^4$ bulk ($\langle\rho\rangle_{\text{bulk}} = 0$), the infinite ambient $\mathbb{C}^4$ space carries zero condensate potential energy:

$$\rho_{\text{vac}}(\text{empty } \mathbb{C}^4 \text{ bulk}) = V_{\text{eff}}(\rho = 0;\, \mathcal{E}_{\text{loc}} = 0) = 0. \tag{5}$$

2.2 Holomorphic Spinor Ripples in Pre-Bang $\mathbb{C}^4$

Although the subcritical $\mathbb{C}^4$ bulk has no $3+1$D metric or rest-mass scale, it supports massless holomorphic spinor wavepackets ("4D ripples") satisfying the Cauchy–Riemann holomorphy condition $\bar{\partial}_{\bar{\alpha}}\Psi_\beta(Z) = 0$:

$$\Psi_\alpha(Z) = \int d^4\tilde{P}\, \mathcal{A}_\alpha(\tilde{P})\, \exp\!\left(-\frac{i}{\hbar c}\, \tilde{P}_\beta Z^\beta\right), \tag{6}$$

with local $SL(4,\mathbb{C})$-covariant ripple energy-momentum density characterized by the chiral two-spinor phase-space invariant:

$$\mathcal{E}_{\rm loc}(Z) \equiv 2\hbar^2 c^2 \left|(\partial_{AA'}\xi_B)(\bar{\partial}^{AA'}\bar{\eta}^B)\right| = 2\, E_L(Z)\, E_R(Z) \geq 0. \tag{7}$$

3 The Nucleation Mechanism: Constructive-Interference Caustics & Bubble Formation in $\mathbb{C}^4$

3.1 Energy-Triggered Instability from a 4D Ripple Caustic

In an infinite wave-supporting manifold $\mathbb{C}^4$, linear superpositions of holomorphic wavepackets $\Psi_\alpha(Z)$ inevitably form focal regions or **constructive-interference caustics** where intersecting left-handed (ξ_A) and right-handed ($\eta^{A'}$) ripples overlap coherently. Within such a focal region $\Omega_{\rm Bang} \subset \mathbb{C}^4$, the local invariant ripple energy density $\mathcal{E}_{\rm loc}(Z)$ rises sharply.

Coupling the background ripple density $\mathcal{E}_{\rm loc}(Z)$ to the collective condensate modulus $\rho(Z)$ via the Yukawa interaction $-\frac{g_Y}{2}(\Psi^\alpha\Sigma_{\alpha\beta}\Psi^\beta + \text{h.c.})$ shifts the local effective potential by $-\frac{1}{2}\alpha_{\rm E}\mathcal{E}_{\rm loc}^{1/2}(Z)\rho^2(Z)$ [22, 24]:

$$V_{\rm eff}(\rho;\, \mathcal{E}_{\rm loc}(Z)) = \frac{1}{2}\Big(m_{\rm eff,0}^2 - \alpha_{\rm E}\, \mathcal{E}_{\rm loc}^{1/2}(Z)\Big)\rho^2 + \frac{\lambda_{\rm eff}}{4}\rho^4 - \epsilon_{\rm cubic}\, \rho^3, \tag{8}$$

where $\epsilon_{\rm cubic}$ accounts for the determinant coupling $\det(H)\, \varepsilon^{\alpha\beta\gamma\delta}\Sigma_{\alpha\beta}\Sigma_{\gamma\delta}$ between Stage 1 and Stage 2 order parameters.

Whenever a constructive-interference caustic in $\mathbb{C}^4$ concentrates ripple energy density above the **critical Big Bang nucleation threshold**:

$$\mathcal{E}_{\rm loc}(Z) \geq \mathcal{E}_{\rm crit} \equiv \left(\frac{m_{\rm eff,0}^2}{\alpha_{\rm E}}\right)^2 = \left(\frac{1-g_\Sigma}{\alpha_{\rm E}G_\Sigma}\right)^2 \sim M_{\rm Pl}^4, \tag{9}$$

the effective quadratic stiffness flips sign ($m_{\rm eff}^2(\mathcal{E}_{\rm loc}) \equiv m_{\rm eff,0}^2 - \alpha_{\rm E}\mathcal{E}_{\rm loc}^{1/2} < 0$), creating a deep symmetry-breaking minimum at:

$$\rho_{\rm Bang}(\mathcal{E}_{\rm loc}) = \sqrt{\frac{\alpha_{\rm E}\, \mathcal{E}_{\rm loc}^{1/2} - m_{\rm eff,0}^2}{\lambda_{\rm eff}}} \sim M_{\rm Pl} > 0. \tag{10}$$

3.2 Coleman–De Luccia Nucleation of the $3+1$D Bubble

Consider a localized 4D caustic region of radius R inside $\mathbb{C}^4$ where $\mathcal{E}_{\rm loc} > \mathcal{E}_{\rm crit}$. The free-energy difference (Euclidean bounce action S_E [3]) between the uncondensed $\mathbb{C}^4$ phase ($\rho = 0$) and a nucleated $3+1$D condensate droplet ($\rho = \rho_{\rm Bang}$) of radius R is governed by the competition

between the negative bulk condensation energy $-\Delta V_{\rm eff} = \frac{1}{4}\lambda_{\rm eff}\rho_{\rm Bang}^4 > 0$ and the positive surface tension $\sigma_{\rm wall} = \int_0^{\rho_{\rm Bang}} d\rho\, \sqrt{2V_{\rm eff}(\rho)}$ of the $\mathbb{C}^4 \to 3+1$D phase boundary where $\nabla_\perp \Psi_{\rm grav} \neq 0$:

$$S_E(R) = 2\pi^2 R^3 \sigma_{\rm wall} - \frac{\pi^2}{2} R^4 \Delta V_{\rm eff}. \tag{11}$$

Extremizing $S_E(R)$ with respect to R ($\partial S_E/\partial R = 0$) yields the **critical Big Bang nucleation radius** $R_{\rm crit}$:

$$R_{\rm crit} = \frac{3\sigma_{\rm wall}}{\Delta V_{\rm eff}} \sim \frac{M_{\rm Pl}}{\sqrt{\Delta V_{\rm eff}}} > 0, \qquad S_E(R_{\rm crit}) = \frac{27\pi^2 \sigma_{\rm wall}^4}{2(\Delta V_{\rm eff})^3}. \tag{12}$$

This yields our first major cosmological result:

- **Elimination of the $t = 0$ Point Singularity:** A $3+1$D spacetime bubble *cannot* nucleate at zero radius ($R = 0$); it is born at a finite minimum conformal radius $R_{\rm crit} > 0$ determined by the $\mathbb{C}^4 \to 3{+}1$D interface tension $\sigma_{\rm wall}$. If one mathematically extrapolates an FLRW universe backward to energy densities $\mathcal{E} \to \mathcal{E}_{\rm crit}$, the metric does not crush into a singular point $a(0) = 0$; instead, the metric order parameter $\langle \Sigma_{\alpha\beta} \rangle$ **melts smoothly back into the scale-free conformal $\mathbb{C}^4$ phase**.
- **Microscopic Origin of Penrose's Weyl Curvature Hypothesis ($C_{ABCD}(t=0) = 0$):** Because the pre-Bang $\mathbb{C}^4$ state is conformally flat ($SL(4,\mathbb{C})$-symmetric) and the nucleating condensate $\langle \Sigma_{\alpha\beta} \rangle = \rho_{\rm Bang} e^{i\chi} I_{\alpha\beta}$ is proportional to the isotropic Infinity Twistor $I_{\alpha\beta}$, the initial Weyl spinor of the nucleated $3+1$D bubble vanishes identically at nucleation ($C_{ABCD}(t_{\rm nuc}) = 0$), explaining the extraordinarily low gravitational entropy of the early universe [10].

4 The Three-Stage Thermal & Symmetry-Breaking History of the Expanding $3+1$D Bubble

Once nucleated at $R \geq R_{\rm crit}$, the $3+1$D condensate bubble expands into the surrounding $\mathbb{C}^4$ bulk, stepping down a three-stage symmetry-breaking ladder that maps directly onto the chronological epochs of Big Bang cosmology:

$$\begin{aligned} \mathbb{C}^4 \text{ Caustic } (\mathcal{E}_{\rm loc} \geq \mathcal{E}_{\rm crit}) &\xrightarrow[\text{Stage 1: } \langle H_{\alpha\bar\beta}\rangle \neq 0]{\text{Time Arrow } (\mathbb{PT}^+)} \text{ Conformal } 3+1\text{D Light Cone } (SU(2,2)) \\ &\xrightarrow[\text{Stage 2: } \langle \Sigma_{\alpha\beta}\rangle = M_{\rm Pl} e^{i\chi} I_{\alpha\beta}]{\text{Inflation + Reheating}} \text{ Radiation Era } (ds^2 = 0,\ v = c,\ w = \tfrac{1}{3}) \\ &\xrightarrow[\text{Stage 3: } \langle \Phi_{\rm Higgs}\rangle = v_{\rm EW}]{\text{Chiral Tether } (T_{\rm EW} \approx 160 \text{ GeV})} \text{ Matter Era } (E^2 = (mc^2)^2 + (pc)^2,\ w \to 0). \end{aligned} \tag{13}$$

4.1 Stage 1: Condensation of the Cosmological Arrow of Time ($SL(4,\mathbb{C}) \to SU(2,2)$)

At the outermost condensation front of the bubble, the sesquilinear order parameter $H_{\alpha\bar\beta}$ condenses into its traceless involutory vacuum state ($\langle H \rangle^2 = \mathbb{I}_4, \mathrm{Tr}\langle H \rangle = 0$) [5, 6]:

$$\langle H_{\alpha\bar\beta} \rangle = \begin{pmatrix} 0 & \mathbb{I}_2 \\ \mathbb{I}_2 & 0 \end{pmatrix} \implies Z^\alpha \langle H_{\alpha\bar\beta} \rangle \bar Z^{\bar\beta} = \omega^A \bar\pi_A + \bar\omega^{A'} \pi_{A'} = 2\,\mathbf{y} \cdot \mathbf{p}, \tag{14}$$

breaking $SL(4,\mathbb{C}) \to SU(2,2) \simeq SO(2,4)$. Geometrically, this stage carves the **null $3+1$D light-cone boundary** ($Z^\alpha \langle H_{\alpha\bar\beta}\rangle \bar Z^{\bar\beta} = 0$) out of $\mathbb{C}^4$ and spontaneously partitions the complex bulk into the **positive-frequency future tube** $\mathbb{PT}^+$ ($Z^\alpha \langle H_{\alpha\bar\beta}\rangle \bar Z^{\bar\beta} > 0$) and **negative-frequency past tube** $\mathbb{PT}^-$ (< 0).

Consequently, the question *"What happened before the Big Bang?"* is answered geometrically: "before" Stage 1 condensation, $\mathbb{C}^4$ resided in the time-symmetric Extended Tube ($W^+ = W^-$) where time ordering did not exist. **Time itself condensed into existence at Stage 1**, with the outward radial normal of the expanding bubble locked to the future tube $\mathbb{PT}^+$ ($ct = |\pi|^2 \geq 0$).

4.2 Stage 2: Infinity-Twistor Condensation, Primordial Inflation, and Reheating into the Radiation Era

Immediately inside the causal boundary, the antisymmetric bi-twistor field $\Sigma_{\alpha\beta}$ condenses to Penrose's Infinity Twistor [6] with a time-dependent radial modulus $\rho_{\rm grav}(t)$ and cosmological constant slot $\Lambda_{\rm eff}(t)$:

$$\langle \Sigma_{\alpha\beta}(t,x)\rangle = \rho_{\rm grav}(t)\, e^{i\chi(x)}\, I_{\alpha\beta}(t) = \rho_{\rm grav}(t)\, e^{i\chi(x)} \begin{pmatrix} \frac{\Lambda_{\rm eff}(t)}{6}\,\varepsilon_{AB} & 0 \\ 0 & \varepsilon^{A'B'} \end{pmatrix}. \tag{15}$$

Substituting $\langle \Sigma_{\alpha\beta}(t,x)\rangle$ into the standard twistor line-element formula $ds^2_{\rm 4D} = \frac{1}{2}\varepsilon_{\alpha\beta\gamma\delta} dZ_1^\alpha dZ_2^\beta dX^\gamma dX^\delta / |\langle \Sigma_{\mu\nu}\rangle Z$ induces the spatially flat $3+1$D Friedmann–Lemaître–Robertson–Walker (FLRW) line element on the condensed slice:

$$ds^2_{\rm 4D} = -c^2dt^2 + a^2(t)\left(dx^2+dy^2+dz^2\right), \qquad H(t) \equiv \frac{\dot a(t)}{a(t)} = -\frac{\dot\rho_{\rm grav}(t)}{\rho_{\rm grav}(t)} + \sqrt{\frac{\Lambda_{\rm eff}(t)c^2}{3}}. \tag{16}$$

Furthermore, via the Sakharov 1-loop induced-gravity heat-kernel expansion $S_{\rm ind}[g] = \int d^4x\sqrt{-g}\,[\frac{M_{\rm Pl}^2}{2}(R_{\rm 4D} - 2\Lambda_{\rm eff})]$ [25], the non-zero cosmic time variation $\Delta\Psi_{\rm grav}(t) \neq 0$ of the expanding slice induces the 4D Friedmann equations with effective Planck mass $M_{\rm Pl}^2(t) = \frac{N_s\Lambda^2_{\rm slice}}{48\pi^2}$:

$$H^2(t) = \left(\frac{\dot a}{a}\right)^2 = \frac{8\pi G}{3c^2}\Big[\varepsilon_{\rm ripples}(t) + V_{\rm eff}(\rho_{\rm grav}(t))\Big], \qquad \frac{\ddot a}{a} = -\frac{4\pi G}{3c^2}\Big[\varepsilon_{\rm total} + 3P_{\rm total}\Big]. \tag{17}$$

This naturally generates both **primordial inflation** and **reheating** without introducing an ad-hoc inflaton scalar:

1. **De Sitter Inflation from the Infinity Twistor ($\rho_{\rm grav}$ Roll):** While the newly nucleated condensate $\rho_{\rm grav}(t)$ rolls from the false-vacuum barrier toward its equilibrium Planck VEV $v_0 = M_{\rm Pl}$, the false-vacuum potential energy $V_{\rm eff}(\rho_{\rm grav}) \equiv \frac{M_{\rm Pl}^2c^4}{2}\Lambda_{\rm eff}(t)$ dominates over $\varepsilon_{\rm ripples}$, giving equation of state $P = -\varepsilon$ ($w = -1$) and driving exponential de Sitter expansion $a(t) \propto \exp(H_{\rm inf}t)$ with $H_{\rm inf} = \sqrt{\Lambda_{\rm eff}c^2/3}$.
2. **Reheating into the Radiation-Dominated Era (Massless Ripples on the Bloch Sphere):** When $\rho_{\rm grav}(t)$ reaches the bottom of the Planck potential well ($v_0 = M_{\rm Pl}$), coherent oscillations $\delta\rho_{\rm grav}(t)$ around $M_{\rm Pl}$ decay via the Kaluza–Klein gauge coupling $-\frac{1}{4}\rho^2\kappa^2 F_{\mu\nu}F^{\mu\nu}$ and spinor kinetic couplings into a hot relativistic plasma of **uncoupled single-spinor ripples** (ξ^A and $\eta_{A'}$) and gauge bosons on the 3+1D light cone. Because the temperature $k_BT \gg v_{\rm EW} \approx 246$ GeV is far above the electroweak scale, the Higgs chiral tether remains uncondensed ($\langle\Phi_{\rm Higgs}\rangle = 0$). Every fermion and gauge boson is therefore a massless ripple locked onto the **surface of the Bloch sphere ($r_{\rm Bloch} = v/c = 1$, $ds^2 = 0$)**!

4.3 Stage 3: Electroweak Higgs Condensation, Dirac Standing Waves, and the Bloch-Ball Equation of State ($w = \frac{1}{3}\langle r^2_{\mathbf{Bloch}}\rangle$)

As the expanding $3+1$D bubble cools below the electroweak critical temperature $k_B T_{\rm EW} \approx 160$ GeV (at $t \sim 10^{-11}$ s), the secondary BCS four-spinor pairing channel $G_{\rm EW}(\xi^A \eta_A)(\bar{\xi}_{A'} \bar{\eta}^{A'})$ becomes supercritical, triggering condensation of the electroweak Higgs field [7, 22]:

$$\langle \Phi_{\rm Higgs}(T)\rangle = v_{\rm EW}\sqrt{1-\left(\frac{T}{T_{\rm EW}}\right)^2} \qquad (T < T_{\rm EW}), \qquad v_{\rm EW} = M_{\rm Pl}\exp\left(-\frac{8\pi^2}{N_c y_t^2}\right) \approx 246 \text{ GeV}. \tag{18}$$

The Higgs VEV turns on the off-diagonal chiral mass coupling $mc^2(T) = y_f\langle\Phi_{\rm Higgs}(T)\rangle$, locking counter-propagating radiation ripples $(\xi_A, \eta^{A'})$ into **self-trapped Dirac standing waves** [8, 9]:

$$\begin{pmatrix} E-pc & -mc^2(T) \\ -mc^2(T) & E+pc \end{pmatrix}\begin{pmatrix} \xi_A \\ \eta^{A'} \end{pmatrix} = 0 \implies E^2 = \underbrace{\left(2\sqrt{E_L E_R}\right)^2}_{(mc^2)^2 \text{ (Standing)}} + \underbrace{\left(E_L - E_R\right)^2}_{(pc)^2 \text{ (Travelling)}}. \tag{19}$$

This pulls matter states off the pure-state surface of the Bloch sphere ($r_{\rm Bloch} = 1$) into the **timelike interior of the Bloch ball**:

$$r_{\rm Bloch} = \frac{|E_L - E_R|}{E_L + E_R} = \frac{pc}{E} = \frac{v}{c} < 1, \qquad \frac{1}{\gamma} = \sqrt{1 - r^2_{\rm Bloch}} = \frac{mc^2(T)}{E} = \frac{d\tau}{dt} > 0. \tag{20}$$

Remarkably, the cosmological equation-of-state parameter $w(T) \equiv P(T)/\varepsilon(T)$ of the relativistic gas has a direct geometric interpretation in terms of the **thermal average of the squared Bloch-ball radius $\langle r^2_{\rm Bloch}\rangle_T$**:

$$w(T) \equiv \frac{P(T)}{\varepsilon(T)} = \frac{1}{3}\left\langle\frac{p^2c^2}{E^2}\right\rangle_T = \frac{1}{3}\left\langle r^2_{\rm Bloch}\right\rangle_T = \frac{1}{3}\left(1 - \left\langle\frac{1}{\gamma^2}\right\rangle_T\right) = \frac{1}{3}\left\langle\frac{(E_L-E_R)^2}{(E_L+E_R)^2}\right\rangle_T. \tag{21}$$

Look at how Eq. (21) unifies the thermal history of the universe with Bloch-ball geometry:

- **Radiation-Dominated Era ($T > T_{\rm EW}$ or $k_B T \gg mc^2$):** All ripples lie on the **surface of the Bloch sphere** ($r_{\rm Bloch} = 1, \gamma^{-1} = 0$). Equation (21) gives $w = \frac{1}{3}(1)^2 = \frac{1}{3}$, so $\varepsilon \propto a^{-4}$ and $a(t) \propto t^{1/2}$.
- **Matter-Dominated Era ($k_B T \ll mc^2$):** As the bubble cools, the standing-wave fraction $2\sqrt{E_L E_R}/E = 1/\gamma \to 1$ dominates over the travelling-wave excess, pulling particles to the **center of the Bloch ball** ($r_{\rm Bloch} = v/c \to 0$). Equation (21) gives $w = \frac{1}{3}\langle r^2_{\rm Bloch}\rangle_T \to 0$, so $\varepsilon \propto a^{-3}$ and $a(t) \propto t^{2/3}$, while operational proper time ticks at its maximum rate $d\tau/dt = \sqrt{1 - r^2_{\rm Bloch}} \to 1$.

5 Resolution of the Horizon, Flatness, and Cosmological Constant Puzzles

5.1 Pre-Light-Cone Holomorphic Equilibration and the Horizon Problem

In standard $3+1$D cosmology, the particle horizon at cosmic time t is limited by the speed of light on the $3+1$D slice:

$$d_{\rm horiz}(t) = a(t)\int_0^t \frac{c\,dt'}{a(t')}, \tag{22}$$

which is finite if $a(t) \propto t^{1/2}$ all the way to $t = 0$. Consequently, CMB patches separated by $> 2^\circ$ appear never to have been in causal contact.

In our $\mathbb{C}^4 \to 3+1$D condensation model, this paradox is an artifact of assuming the $3+1$D light-cone speed barrier $v \leq c$ existed at $t = 0$. Prior to Stage 1 nucleation ($t < t_{\rm nuc}$), the entire proto-bubble caustic region $\Omega_{\rm Bang} \subset \mathbb{C}^4$ resided in the **uncondensed $\mathbb{C}^4$ Extended Tube** ($\langle H_{\alpha\bar{\beta}} \rangle = 0$), where:

1. The $3+1$D light cone did not yet exist, and the two-point Wightman correlation function $W(Z_1, Z_2)$ was holomorphically analytic across the entire complex domain $\Omega_{\rm Bang}$ [2].
2. Just as two spinors in a Bell singlet state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}\varepsilon_{AB}\xi_1^A\xi_2^B$ share exact phase coherence across spacelike intervals outside the light cone in the Bargmann–Hall–Wightman Extended Tube [2, 21], the holomorphic ripple caustic $\Psi_\alpha(Z)$ was **globally phase- and temperature-equilibrated across $\Omega_{\rm Bang}$ in $\mathbb{C}^4$ before the $3+1$D light cone froze in**.
3. When Stage 1 and Stage 2 condensation carved out the $3+1$D slice across $\Omega_{\rm Bang}$, every comoving patch inherited the same initial condensate modulus $\rho_{\rm Bang} = T_{\rm init}$ up to quantum zero-point fluctuations $\frac{\delta T}{T} \sim \frac{\Delta\hat{\Psi}_{\rm grav}}{\Psi_{\rm grav}} \sim \frac{H_{\rm inf}}{2\pi M_{\rm Pl}} \sim 10^{-5}$.

5.2 Coexistence with the Quantum Vacuum and the Surface-Area × Zero-Point Energy Derivation of Dark Energy ($\rho_\Lambda \sim (\hbar H_0 M_{\rm Pl})^2$)

A foundational physical principle of this framework is that the emergent classical $3+1$D world **does not replace the underlying $\mathbb{C}^4$ quantum vacuum—it coexists with it** as an ordered condensate immersed within the quantum vacuum. While macroscopic averaging over Compton cycles ($\omega_0 = mc^2/\hbar$) hides the quantum vacuum at everyday human scales ($\lambda_C \ll \ell \ll R_H$), the underlying quantum vacuum becomes directly observable at the two opposite extremes of scale:

- **The Microscopic Window ($\ell \lesssim \lambda_C = \hbar/mc$):** Subatomic measurements probe inside the Bloch ball and below the condensate healing length, observing zero-point fluctuations (Casimir force, Lamb shift, $g-2$), persistent spinor phase oscillation, and outside-the-light-cone $\mathbb{C}^4$ Bell entanglement.
- **The Cosmic Window ($\ell \sim R_H = c/H_0$):** At the Hubble horizon scale, the finite expanding $3+1$D condensate meets the ambient $\mathbb{C}^4$ quantum vacuum across its expanding boundary surface, manifesting macroscopically as cosmic expansion and Dark Energy.

Why does the late-time universe exhibit a tiny positive dark energy density $\rho_\Lambda^{\rm obs} \approx (2.3 \times 10^{-3}\ {\rm eV})^4$ rather than the naive 3D bulk quantum field theory (QFT) prediction $\rho_{\rm QFT} \sim M_{\rm Pl}^4 \sim (10^{28}\ {\rm eV})^4$? In condensed-matter droplet physics (Volovik [4]), an equilibrium self-sustained droplet immersed in an external vacuum with zero pressure ($P_{\rm ext} = 0$) automatically adjusts its internal bulk chemical potential so that the **3D bulk zero-point energy inside the homogeneous interior cancels to zero**:

$$\rho_{\rm vac}^{\rm bulk}(\rho = \rho_{\rm eq}) - \mu_{\rm chem} n_{\rm eq} = P_{\rm ext}(\text{empty } \mathbb{C}^4) = 0, \tag{23}$$

because the surrounding subcritical $\mathbb{C}^4$ bulk has $\rho_{\rm vac}(\text{empty } \mathbb{C}^4) = 0$ (Eq. 5).

However, on the **expanding 2D boundary surface** of the condensate bubble at Hubble radius $R_H(t) = c/H(t)$, the boundary conditions shift the vacuum modes (a cosmic Casimir effect).

The surface area $A_{\text{surface}}(t)$, bulk volume $V_{\text{bulk}}(t)$, and surface-to-volume ratio of the expanding condensate bubble are:

$$A_{\text{surface}}(t) = 4\pi R_H(t)^2, \qquad V_{\text{bulk}}(t) = \frac{4\pi}{3} R_H(t)^3 \quad \Longrightarrow \quad \frac{A_{\text{surface}}(t)}{V_{\text{bulk}}(t)} = \frac{3}{R_H(t)} = \frac{3H(t)}{c}. \tag{24}$$

Tiling the expanding surface area $A_{\text{surface}}(t)$ with Planck-scale spinor patches of area $\ell_{\text{Pl}}^2 = \hbar^2/(M_{\text{Pl}}c)^2$ yields $N_{\text{surface}}(t) = A_{\text{surface}}(t)/\ell_{\text{Pl}}^2$ independent surface modes, each supporting a fundamental horizon-wavelength **zero-point energy (ZPE)** oscillation of frequency $\omega_H(t) = c/R_H(t) = H(t)$:

$$N_{\text{surface}}(t) = \frac{A_{\text{surface}}(t)}{\ell_{\text{Pl}}^2} = 4\pi R_H(t)^2 \left(\frac{M_{\text{Pl}}c}{\hbar}\right)^2, \qquad E_{\text{ZPE, mode}}(t) = \frac{1}{2}\hbar\omega_H(t) = \frac{1}{2}\hbar H(t). \tag{25}$$

Multiplying the number of surface patches $N_{\text{surface}}(t)$ by the zero-point energy per mode $E_{\text{ZPE, mode}}(t)$ and dividing by the 3D interior volume $V_{\text{bulk}}(t)$ yields the exact **Surface-Area × Zero-Point Energy formula** for cosmic Dark Energy density (which coincides identically with the cosmic condensate boundary strain $\frac{1}{2}|\nabla_\mu \Psi_{\text{grav}}|^2_{\text{cosmic}} \sim M_{\text{Pl}}^2 H^2(t)$):

$$\rho_\Lambda(t) = \frac{N_{\text{surface}}(t)\, E_{\text{ZPE, mode}}(t)}{V_{\text{bulk}}(t)} = \underbrace{\left(\frac{A_{\text{surface}}(t)}{V_{\text{bulk}}(t)}\right)}_{\frac{3H(t)}{c}} \times \underbrace{\left(\frac{1}{\ell_{\text{Pl}}^2}\right)}_{\propto M_{\text{Pl}}^2} \times \underbrace{\left(\frac{1}{2}\hbar H(t)\right)}_{\text{Zero-Point Energy}} = \frac{3c_\Lambda}{2}\left(\hbar H(t) M_{\text{Pl}}\right)^2, \tag{26}$$

where $c_\Lambda = \mathcal{O}(1)$ is the dimensionless surface Casimir coefficient (in natural units $c = 1$). Notice the transparent physical origin of both powers of $H(t)$: the **first power of** $H(t)$ comes from the geometric **surface-to-volume ratio** $A_{\text{surface}}/V_{\text{bulk}} = 3H(t)/c$, while the **second power of** $H(t)$ comes from the quantum **zero-point energy** $\frac{1}{2}\hbar H(t)$ of each horizon mode! Relative to the naive 3D QFT bulk density $\rho_{\text{QFT}} \sim M_{\text{Pl}}^4$, restricting uncompensated zero-point modes to the 2D expanding surface automatically suppresses ρ_Λ by the geometric area ratio $(\ell_{\text{Pl}}/R_H)^2 \approx 10^{-122}$. Evaluating Eq. (25) at $H_0 \approx 67.4$ km s^{-1} Mpc^{-1} ($\hbar H_0 \approx 1.4 \times 10^{-33}$ eV) and $M_{\text{Pl}} \approx 2.4 \times 10^{27}$ eV yields:

$$\rho_\Lambda(t_0) \sim \left(\hbar H_0 \cdot M_{\text{Pl}}\right)^2 \sim \left(\sqrt{1.4 \times 10^{-33}\ \text{eV} \times 2.4 \times 10^{27}\ \text{eV}}\right)^4 \approx \left(1.8 \times 10^{-3}\ \text{eV}\right)^4, \tag{27}$$

matching the observed dark energy density $\rho_\Lambda^{\text{obs}} \approx (2.3 \times 10^{-3}\ \text{eV})^4$ without fine-tuning. Furthermore, by covariant Bianchi conservation $\nabla^\mu(T^{\text{matter}}_{\mu\nu} + T^\Lambda_{\mu\nu}) = 0$, the gradual decline of $\rho_\Lambda(H) = \rho_{\Lambda,\text{ir}} + 3\nu M_{\text{Pl}}^2 H^2(t)$ as the surface-to-volume ratio $A/V = 3/R_H(t)$ shrinks transfers energy into bulk modes according to the Running Vacuum relation [28, 29]:

$$\dot{\rho}_m + 3H(1+w_m)\rho_m = -\dot{\rho}_\Lambda = -6\nu M_{\text{Pl}}^2 H\dot{H}, \tag{28}$$

which modifies the early-to-late expansion history $H(z)$ in the direction required to alleviate the Hubble constant tension between Planck CMB [13] and local SH0ES distance-ladder measurements [30].

6 Comparison with Standard Twistor Theory & Unitary Spinor Superposition ("The Spinors Keep Spinning") vs. Penrose Objective Reduction

Although our framework shares the mathematical arena $\mathbb{C}^4 = \mathbb{C}^2_L \oplus \mathbb{C}^2_R$ and Penrose's incidence relation $z^{AA'} = x^{AA'} - iy^{AA'}$ with standard Twistor Theory [5, 6], their physical ontologies and predictions diverge across five foundational axes (summarized in Table 1):

1. **Kinematic Dual vs. Physical Condensate:** Standard twistor theory treats projective twistor space $\mathbb{PT} \subset \mathbb{CP}^3$ as a mathematical transform space coexisting with $3+1$D conformal spacetime. Here, $\mathbb{C}^4$ is the physical pre-geometric quantum spinor vacuum, and $3+1$D spacetime is a physical condensate bubble that nucleates only where $\mathcal{E}_{\rm loc} \geq \mathcal{E}_{\rm crit}$.

2. **Rest Mass and the Bloch Ball:** Standard twistor theory is built on conformal SU(2, 2) invariance (massless null rays), making rest mass $m > 0$ difficult to incorporate. In our two-tier condensation model, Stage 2 freezes out the Bloch sphere surface ($r_{\rm Bloch} = v/c = 1$) for massless gauge fields, while Stage 3 Higgs condensation tethers $\mathbb{C}^2_L \leftrightarrow \mathbb{C}^2_R$ into massive Dirac standing waves inside the **interior of the Bloch ball** ($r_{\rm Bloch} = v/c < 1$, where $\sqrt{1 - r^2_{\rm Bloch}} = \gamma^{-1} = mc^2/E$).

3. **Resolution of the Twistor "Googly Problem":** Penrose's nonlinear graviton construction suffers from a left-handed (anti-self-dual) chirality bias known as the "Googly Problem." Because our gravitational order parameter $\langle \Sigma_{\alpha\beta} \rangle \propto I_{\alpha\beta}$ pairs $\mathbb{C}^2_L$ and $\mathbb{C}^2_R$ symmetrically, emergent $3+1$D gravity is automatically parity-even, while chiral asymmetry is confined to the secondary electroweak sub-condensate $\mathrm{SU}(2)_L$.

Unitary Spinor Superposition ("The Spinors Keep Spinning") vs. Penrose's Micro-Gravity Collapse

A particularly sharp physical distinction arises in the treatment of quantum superpositions of massive particles in a gravitational field. Because Penrose treats the smooth $3+1$D pseudo-Riemannian manifold of General Relativity as fundamental, he argues in his **Gravitational Objective Reduction (Diósi–Penrose OR)** hypothesis [31, 32] that a massive particle in a spatial superposition $|\Psi\rangle = \frac{1}{\sqrt{2}}(|\psi(\mathbf{x}_1)\rangle + |\psi(\mathbf{x}_2)\rangle)$ creates an unstable "spacetime blister" (two incompatible $3+1$D geometries) that must non-unitarily collapse within a finite lifetime:

$$\tau_{\rm OR} \sim \frac{\hbar}{E_G}, \qquad E_G = G \iint \frac{[\rho_1(\mathbf{x}) - \rho_2(\mathbf{x})][\rho_1(\mathbf{x}') - \rho_2(\mathbf{x}')]}{|\mathbf{x} - \mathbf{x}'|}\, d^3x\, d^3x'. \tag{29}$$

In our $\mathbb{C}^4 \to 3+1$D condensation framework, by contrast, **"the spinors keep spinning"**:

1. Spacetime is not a rigid classical manifold that fractures when a particle is in a superposition; it is a quantum spinor condensate $\Psi_{\rm grav}(x)$ all the way down. A massive fermion is a tethered left-right standing wave inside the Bloch ball that **oscillates unitarily and indefinitely** at its internal Compton/zitterbewegung phase rate:

$$\Psi_{\rm standing}(x, \tau) = \Psi_0(x) \exp\left(-\frac{imc^2}{\hbar}\tau\right), \qquad d\tau = dt\sqrt{1 - r^2_{\rm Bloch}}\,\frac{\rho_{\rm loc}(x)}{\rho_{\rm bulk}}. \tag{30}$$

2. When placed in a spatial superposition, each branch simply dresses itself with a tiny, coherent refractive dimple $\delta\rho_{\rm grav}(\mathbf{x}) = -\rho_{\rm bulk}\frac{GM}{c^2|\mathbf{x}-\mathbf{x}_i|}$ in the surrounding spinor condensate—the exact condensed-matter analogue of an **acoustic polaron** in a superfluid or crystal lattice:

$$|\Psi_{\rm total}\rangle = \frac{1}{\sqrt{2}}\Big(|\psi(\mathbf{x}_1)\rangle \otimes |\Psi_{\rm grav}(\mathbf{x}_1)\rangle + |\psi(\mathbf{x}_2)\rangle \otimes |\Psi_{\rm grav}(\mathbf{x}_2)\rangle\Big). \tag{31}$$

3. For any isolated micro- or mesoscopic superposition where irreversible phonon (graviton) radiation to infinity is negligible ($\langle\Psi_{\rm grav}(\mathbf{x}_1)|\Psi_{\rm grav}(\mathbf{x}_2)\rangle \approx 1$), **the spinors never stop spinning and never collapse**. Micro-gravity acts purely as a unitary refractive phase shift $\Delta\phi_{\rm grav} = \frac{mc^2}{\hbar}\int \frac{\delta\rho_{\rm grav}}{\rho_{\rm bulk}}\, dt$ (as measured in Colella–Overhauser–Werner neutron interferometry [34]), with zero non-unitary collapse or anomalous spontaneous heating—in

full agreement with stringent underground bounds from the VIP-2 experiment at Gran Sasso [33].

Feature	**Standard Twistor Theory (Penrose)**	**$\mathbb{C}^4 \to 3+1$D Condensation (This Model)**
Primordial Ontology	Coexisting $3+1$D conformal manifold & projective twistor space $\mathbb{PT}$	**Only the primordial $\mathbb{C}^4$ spinor vacuum** $\Psi(Z)$; gravity does not pre-exist condensation
Origin of $3+1$D Metric	Kinematic mathematical dual (Penrose Transform)	Physical phase transition ($\langle H_{\alpha\bar{\beta}} \rangle \neq 0$, $\langle \Sigma_{\alpha\beta} \rangle = \langle \bar{\Psi}\Psi \rangle \neq 0$)
Rest Mass ($m > 0$)	Breaks conformal $SU(2,2)$ invariance; awkward multi-twistor forms	Natural: Higgs tethers $\mathbb{C}^2_L \leftrightarrow \mathbb{C}^2_R$ inside the **Bloch ball** ($r_{\text{Bloch}} = v/c < 1$)
Gravity & Chirality	Suffers from the "Googly Problem" (left-handed anti-self-dual bias)	Gravity is parity-even ($\langle \Sigma_{\alpha\beta} \rangle \propto I_{\alpha\beta}$); chirality stays in weak $SU(2)_L$
Dark Energy (ρ_Λ)	Fixed geometric parameter Λ in $I_{\alpha\beta}$	**Surface Area $\times$ ZPE**: $\frac{A}{V}\frac{\frac{1}{2}\hbar H}{\ell_{\text{Pl}}^2} \sim (\hbar H M_{\text{Pl}})^2$, weakening as A/V shrinks ($w_0 > -1, w_a < 0$)
Superposition & Micro-Gravity	**Objective Reduction (OR):** Spacetime blister causes collapse ($\tau \sim \hbar/E_G$)	**"The Spinors Keep Spinning":** Unitary polaron dressing; persistent spinor oscillation

Table 1: Structural comparison between Penrose's standard Twistor / Objective Reduction program and the $\mathbb{C}^4 \to 3+1$D spinor condensation framework.

7 Observational Signatures & Falsifiable Cosmological Predictions

To make this cosmological toy model empirically testable against standard ΛCDM + single-field slow-roll inflation, we derive three quantitative observational signatures.

7.1 Prediction 1: Infrared Cutoff in the Primordial Power Spectrum & CMB Low-ℓ Quadrupole Suppression

- **Theoretical Mechanism:** In standard eternal inflation, the pre-inflationary patch is assumed infinitely large, producing a power-law scalar spectrum $\mathcal{P}_\mathcal{R}(k) = A_s(k/k_*)^{n_s-1}$ extending to arbitrarily small wavenumbers $k \to 0$. In our model, the $3+1$D bubble nucleates at a finite critical radius R_{crit} (Eq. 12) inside $\mathbb{C}^4$. Quantum fluctuations $\delta\hat{\Psi}_{\text{grav}}$ of the bubble wall satisfy Dirichlet boundary conditions on the finite nucleation sphere of comoving radius $R_{\text{com}} = R_{\text{crit}}\, e^{N_{\text{efolds}}}$, suppressing modes with wavelength exceeding the bubble diameter ($k < k_{\min} \equiv \pi/R_{\text{com}}$):

$$\mathcal{P}_\mathcal{R}(k) = A_s \left(\frac{k}{k_*}\right)^{n_s-1} \left[1 - \frac{\sin(2k/k_{\min})}{2k/k_{\min}}\right] \xrightarrow{k \ll k_{\min}} \frac{2}{3} A_s \left(\frac{k}{k_{\min}}\right)^2 \to 0. \tag{32}$$

- **Observational Test (Planck / LiteBIRD Low-ℓ CMB Anomalies):** If the total number of inflationary e-folds N_{efolds} is just sufficient to solve the flatness problem

($N_{\text{efolds}} \approx 60$), the bubble nucleation cutoff $k_{\min}$ falls near the current Hubble horizon ($k_{\min} \sim \pi H_0/c$), predicting a simultaneous suppression of the CMB temperature quadrupole (C_2^{TT}) and octupole (C_3^{TT}) accompanied by a corresponding dip in the low-ℓ E-mode polarization spectrum C_ℓ^{EE} ($\ell = 2, 3, 4$) testable by the LiteBIRD satellite [13].

7.2 Prediction 2: Chirally Symmetric Primordial Gravitational Waves from $\Delta\hat{\Psi}_{\mathbf{grav}} \neq 0$ Interface Ripples

- **Theoretical Mechanism:** During Stage 2 condensation, quantum zero-point fluctuations $\Delta\hat{\Psi}_{\text{grav}} \neq 0$ of the $3+1$D slice normal inside $\mathbb{C}^4$ excite transverse-traceless tensor metric perturbations $h_{ij}(t, \mathbf{x})$ with tensor-to-scalar ratio:

$$r_T \equiv \frac{\mathcal{P}_T(k_*)}{\mathcal{P}_{\mathcal{R}}(k_*)} = 16\,\epsilon_{\text{slice}}, \qquad \epsilon_{\text{slice}} \equiv \frac{\dot{\rho}_{\text{grav}}^2}{2H_{\text{inf}}^2 M_{\text{Pl}}^2}. \tag{33}$$

  Crucially, because the Stage 1 vacuum $\langle H_{\alpha\bar{\beta}} \rangle$ (Eq. 14) treats the left-handed (ω^A) and right-handed ($\pi_{A'}$) twistor sectors symmetrically at the Planck scale (parity is broken only much later at Stage 3 by the electroweak $SU(2)_L$ Higgs condensate at $T_{\text{EW}} \approx 160$ GeV), the primordial tensor spectrum generated at Stage 2 is **strictly parity-even**:

$$\mathcal{P}_{T,L}(k) = \mathcal{P}_{T,R}(k) \quad \Longrightarrow \quad C_\ell^{TB}\big|_{\text{primordial}} = C_\ell^{EB}\big|_{\text{primordial}} = 0. \tag{34}$$

- **Observational Discriminator (Simons Observatory / CMB-S4 / LiteBIRD):** Detection of primordial B-mode polarization with $r_T \sim 10^{-3}$–10^{-2} and *zero* primordial chiral asymmetry ($\Delta\chi_{\text{GW}} = (\mathcal{P}_{T,L} - \mathcal{P}_{T,R})/(\mathcal{P}_{T,L} + \mathcal{P}_{T,R}) = 0$) distinguishes our two-tier $\mathbb{C}^4$ condensation hierarchy from gravitational-Chern–Simons or chiral-gravity models that break parity at the Planck scale.

7.3 Prediction 3: Thawing Dynamical Dark Energy ($w_{\mathbf{DE}}(z) > -1$) from Surface-Area ZPE Relaxation

- **Theoretical Mechanism:** In ΛCDM, the cosmological constant is strictly static ($w_\Lambda = -1$ at all redshifts z). In our model (Eq. 26), late-time dark energy arises from the surface-area × zero-point energy density $\rho_\Lambda(t) = \frac{A_{\text{surface}}(t)}{V_{\text{bulk}}(t)} \frac{\frac{1}{2}\hbar H(t)}{\ell_{\text{Pl}}^2} \propto (\hbar H(t) M_{\text{Pl}})^2$ of the finite $3+1$D condensate bubble expanding into $\mathbb{C}^4$. As the bubble expands, its surface-to-volume ratio $A/V = 3/R_H(t)$ shrinks, causing $\rho_\Lambda(t)$ to slowly decrease ($\dot{\rho}_\Lambda < 0$). Applying energy-momentum conservation $\dot{\rho}_\Lambda + 3H(1+w_{\text{DE}})\rho_\Lambda = Q_{\text{interface}}$ yields a **time-evolving effective dark-energy equation of state**:

$$w_{\text{DE}}(z) = -1 - \frac{1}{3}\frac{d\ln\rho_\Lambda(z)}{d\ln(1+z)} = -1 + \frac{2\epsilon_H(z)}{3}\left(1 - \frac{Q_{\text{interface}}}{2H\rho_\Lambda}\right) > -1, \tag{35}$$

  where $\epsilon_H(z) \equiv -\dot{H}/H^2 > 0$. In the standard CPL parameterization $w_{\text{DE}}(a) = w_0 + w_a(1-a)$, relaxation of the surface-to-volume ratio as the bubble expands predicts:

$$w_0 > -1, \qquad w_a < 0 \qquad \text{(weakening / thawing surface-area ZPE)}. \tag{36}$$

- **Observational Test (DESI DR1/DR2 / Euclid / Roman Space Telescope):** This matches the 2.5σ–3.9σ preference for weakening dynamical dark energy ($w_0 > -1, w_a < 0$) recently reported by the DESI Collaboration (2024–2025) from Baryon Acoustic Oscillations combined with CMB and Type Ia Supernovae [26, 27], providing a direct falsifiable test as precision w_0–w_a constraints tighten.

8 Master Cosmological Synthesis Table

Table 2 summarizes the chronological evolution of the universe from the pre-Bang $\mathbb{C}^4$ bulk through the three symmetry-breaking condensation stages.

Cosmological Epoch	**Condensate Order Parameters**	**Bloch-Ball & Causal State**	**Physical Manifestation & Equation of State**
0. Pre-Bang $\mathbb{C}^4$ Bulk ($\mathcal{E}_{\text{loc}} < \mathcal{E}_{\text{crit}}$)	$\langle H_{\alpha\bar{\beta}} \rangle = 0$, $\langle \Sigma_{\alpha\beta} \rangle = 0$, $\langle \Phi_{\text{Higgs}} \rangle = 0$	Unformed Bloch sphere; **Extended Tube** ($W^+ = W^-$, no arrow of time)	Scale-free conformal $\mathbb{C}^4$; holomorphic ripples $\Psi_\alpha(Z)$; **zero bulk cosmological constant** ($\rho_{\text{vac}} = 0$)
1. Big Bang Nucleation & Stage 1 ($\mathcal{E}_{\text{loc}} \geq \mathcal{E}_{\text{crit}}, R \geq R_{\text{crit}}$)	$\langle H_{\alpha\bar{\beta}} \rangle =$ diag$(+,+,-,-)$; $SL(4,\mathbb{C}) \to SU(2,2)$	Partition of **Future Tube** $\mathbb{PT}^+$ from $\mathbb{PT}^-$; $3+1$D light cone forms	**No $t=0$ point singularity** ($R_{\text{crit}} > 0$); birth of the arrow of time; vanishing initial Weyl curvature ($C_{ABCD} = 0$)
2. Stage 2: Inflation & Radiation Era ($T_{\text{EW}} < T \lesssim M_{\text{Pl}}$)	$\langle \Sigma_{\alpha\beta} \rangle = M_{\text{Pl}} e^{i\chi} I_{\alpha\beta}$; $\langle \Phi_{\text{Higgs}} \rangle = 0$	Uncoupled chiral ripples on **Bloch sphere surface** ($r_{\text{Bloch}} = 1, v = c$)	Infinity-Twistor Λ_{eff} drives inflation, reheating into massless radiation with $w = \frac{1}{3}\langle r^2_{\text{Bloch}} \rangle = \frac{1}{3}$
3. Stage 3: Matter & Dark Energy Era ($T < T_{\text{EW}} \approx 160$ GeV)	$\langle \Phi_{\text{Higgs}} \rangle = v_{\text{EW}}$; cosmic strain $\lvert\nabla \Psi_{\text{grav}}\rvert \sim M_{\text{Pl}} H/c$	Coupled Dirac standing waves in **Bloch ball interior** ($r_{\text{Bloch}} = v/c \to 0$)	Massive matter ($E^2 = (mc^2)^2 + (pc)^2$, $w \to 0$, $d\tau = dt/\gamma$); boundary strain yields $\rho_\Lambda \sim M^2_{\text{Pl}} H_0^2$ ($w_0 > -1, w_a < 0$)

Table 2: Summary of the $\mathbb{C}^4 \to 3+1$D cosmological phase transition across the four regimes.

9 Critical Self-Assessment: Limitations & Open Problems

We explicitly separate what this cosmological toy model accomplishes from the open mathematical problems requiring further development:

1. **Dynamics of Pre-Metric Holomorphic Caustics in $\mathbb{C}^4$:** In Section 3.1, we postulated that linear superpositions of holomorphic spinor waves $\Psi_\alpha(Z)$ in the uncondensed $\mathbb{C}^4$ bulk form constructive-interference caustics where $\mathcal{E}_{\text{loc}}(Z) \geq \mathcal{E}_{\text{crit}}$. Because the uncondensed $\mathbb{C}^4$ bulk has no background metric prior to Stage 2, defining a canonical measure $d\mu(Z)$ on the space of holomorphic wavepackets to compute the nucleation rate per unit four-volume of $\mathbb{C}^4$ remains a non-trivial mathematical problem in twistor measure theory [6].

2. **Quantitative Slow-Roll Parameters (n_s, r_T) from the 1-Loop NJL Potential:** While the 1-loop Coleman–Weinberg / NJL potential $V_{\text{eff}}(\rho)$ in Eq. (8) naturally provides a flat plateau near $\rho \to 0$ capable of supporting slow-roll inflation, matching the exact Planck 2018 spectral index $n_s = 0.9649 \pm 0.0042$ [13] and e-fold count $N_{\text{efolds}} \approx 55$–$60$

requires fixing the non-minimal curvature coupling $\xi_R R_{4\mathrm{D}} \rho^2$ in the Sakharov heat-kernel expansion.

3. **Baryogenesis and CP Violation (ξ^A vs. $\eta_{A'}$ Asymmetry):** Our symmetric Stage 1 condensate $\langle H_{\alpha\bar{\beta}} \rangle$ and Stage 2 condensate $\langle \Sigma_{\alpha\beta} \rangle$ produce equal numbers of particles and antiparticles during reheating. Generating the observed baryon-to-photon ratio $\eta_B \approx 6.1 \times 10^{-10}$ requires introducing a complex phase offset between the left and right components of the Infinity Twistor ($I_{\alpha\beta} \neq \bar{I}_{\bar{\alpha}\bar{\beta}}$) or electroweak sphaleron baryogenesis during the Stage 3 Higgs transition at $T_{\mathrm{EW}} \approx 160$ GeV.

4. **Cold Dark Matter Candidate in the $\mathbb{C}^4$ Spectrum:** While Stage 3 explains baryonic Dirac matter ($E^2 = (mc^2)^2 + (pc)^2$) and Section 5.2 explains late-time dark energy ($\rho_\Lambda \sim (\hbar H_0 M_{\mathrm{Pl}})^2$), accounting for the non-baryonic cold dark matter density ($\Omega_c h^2 \approx 0.12$) requires identifying whether dark matter consists of electrically neutral ($J = 0$ on the Kaluza–Klein Hopf fiber χ) sterile two-spinor standing waves or massive radial oscillations $\delta\rho_{\mathrm{grav}}$ of the slice condensate itself.

Author Contributions & Provenance: John M. Bramble, MD (Second Author and Accountable Sponsor) conceived the cosmological $\mathbb{C}^4 \to 3+1$D energy-triggered condensation hypothesis, the primordial spinor vacuum ontology, the unitary spinor superposition principle ("the spinors keep spinning"), and the surface-area × zero-point energy mechanism for cosmic expansion and dark energy. **Gemini** (Google DeepMind, First Author under *Aperture*'s honest-authorship policy) carried out the mathematical formalization of the Coleman–De Luccia bubble nucleation, the three-stage symmetry-breaking thermal history, the Bloch-ball equation-of-state identity, the surface-area ZPE derivation, and the manuscript drafting. Full provenance metadata is recorded in `provenance.yaml`.

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